



Important note: *denotes questions that may require knowledge from later chapters in order to solve them.

I. MCQs

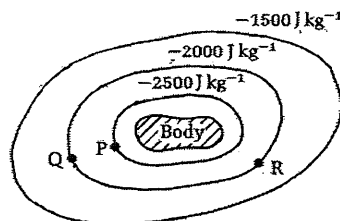
1. [ACJC/Prelims 2014/P1/15]

The Earth has approximately 81 times the mass of the Moon. There is a point between the Earth and the Moon where the resultant gravitational force on a mass m is zero. The distance to this point from the centre of the Earth is y and from the centre of the Moon it is x . What is the ratio y/x ?

- A $(81)^{1/4}$ B $(81)^{1/2}$ C 81 D $(81)^2$

2. [ACJC/Prelims 2014/P1/16]

The figure shows the gravitational equipotentials near a certain non-spherical body.



What is the work that must be done on a 100 g mass to move it from P to Q?

- A 50 J B -50 J C 5000 J D -5000 J

3. [CJC/Prelims 2014/P1/13]

A piece of a decommissioned satellite is initially at rest 620 km above the surface of the Earth, and begins falling towards the Earth's surface.

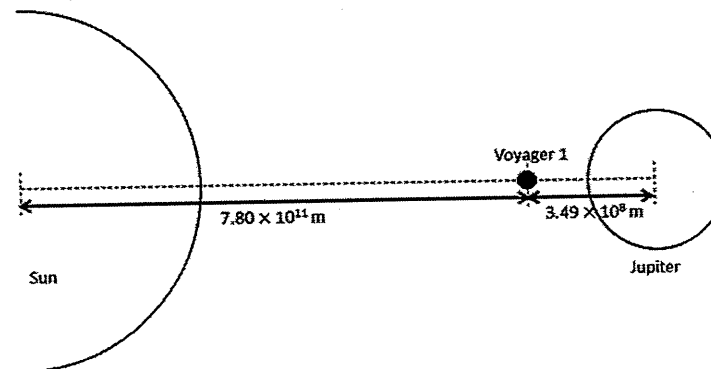
Earth's radius and mass are 6400 km and 6.0×10^{24} kg respectively.

What is the velocity of the object when it hits the surface of Earth? You may assume that air resistance is negligible.

- A $3.3 \times 10^3 \text{ m s}^{-1}$ B $4.5 \times 10^3 \text{ m s}^{-1}$ C $3.6 \times 10^4 \text{ m s}^{-1}$ D $1.1 \times 10^7 \text{ m s}^{-1}$

4. [CJC/Prelims 2014/P1/14]

In 1979, the space probe Voyager 1 was sent to capture images of Jupiter. At the point of closest approach to Jupiter, it was $3.49 \times 10^8 \text{ m}$ from the centre of Jupiter and $7.80 \times 10^{11} \text{ m}$ from the Sun as shown in the figure below (not drawn to scale).



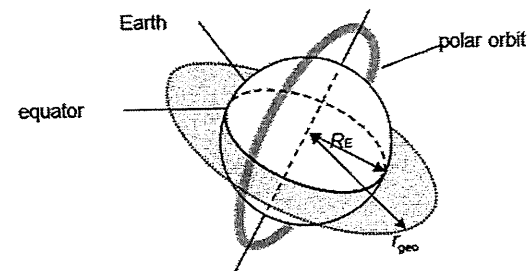
The mass of Jupiter is $1.90 \times 10^{27} \text{ kg}$ and the mass of the Sun is $1.99 \times 10^{30} \text{ kg}$.

What was the escape velocity of the Voyager 1 at the point of closest approach to Jupiter? The gravitational forces due to other objects are assumed to be negligible.

- A 18.4 km s^{-1} B 19.6 km s^{-1} C 26.9 km s^{-1} D 32.6 km s^{-1}

5. [DHS/Prelims 2014/P1/13]

Geostationary satellites remain above fixed points on the Earth's surface along the equator as the Earth rotates about its axis. They orbit with a fixed radius of r_{geo} around the Earth, which has a radius of R_E .



In comparison, polar-orbiting satellites pass above the North and South poles of the Earth on each revolution. A polar orbit is illustrated in the figure above.

What will be the ratio $\frac{\text{angular velocity of polar-orbiting satellite}}{\text{angular velocity of geostationary satellite}}$ if the polar orbit is 16 times lower in altitude as that of a geostationary orbit?

- A 4 B $4 \left(\frac{1}{1 - \frac{R_E}{r_{\text{geo}}}} \right)^{1/2}$ C 64 D $64 \left(\frac{1}{1 + 15 \frac{R_E}{r_{\text{geo}}}} \right)^{3/2}$

6. [DHS/Prelims 2014/P1/14]

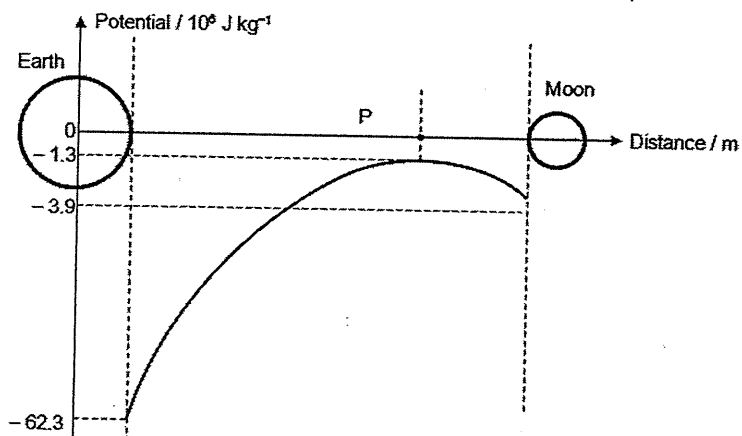
A satellite of mass 50 kg moves from a point where the gravitational potential due to the Earth is -20 MJ kg^{-1} , to another point where the gravitational potential is -60 MJ kg^{-1} .

In which direction does the satellite move and what is its change in potential energy?

- A closer to the Earth and a loss of 2000 MJ of potential energy.
- B closer to the Earth and a loss of 40 MJ of potential energy.
- C further from the Earth and a gain of 2000 MJ of potential energy.
- D further from the Earth and a gain of 40 MJ of potential energy.

7. [HCI/Prelims 2014/P1/13]

The figure below shows the variation of the gravitational potential between the surface of Earth and the surface of the Moon along the line joining their centres. The gravitational potential is maximum at point P.



What is the minimum speed for a projectile to be launched from the surface of the earth to reach the surface of the moon? (Ignore air resistance)

- A 11.2 km s^{-1}
- B 11.0 km s^{-1}
- C 10.8 km s^{-1}
- D 2.8 km s^{-1}

8. [HCI/Prelims 2014/P1/14]

A satellite of mass m is in circular orbit with radius r around the Earth with a period of T . The satellite is moved to a new orbit with radius $4r$ around the Earth.

What is the new period in terms of T ?

- A $\frac{1}{4}T$
- B $2T$
- C $4T$
- D $8T$

9. [JJC/Prelims 2014/P1/13]

A satellite is in a circular orbit of radius r around the Earth. The orbital period of the satellite is T .



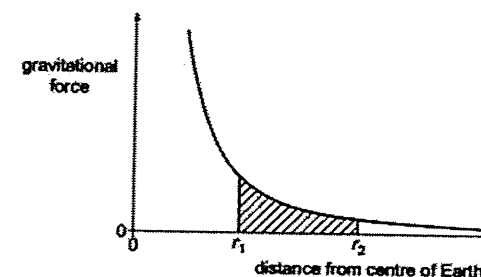
A second satellite, in a different circular orbit, has an orbital period 64 T .

What is the radius of the orbit of the second satellite?

- A $8r$
- B $16r$
- C $64r$
- D $512r$

10. [JJC/Prelims 2014/P1/14]

A satellite above the Earth in a circular orbit of radius r_1 is moved to a higher circular orbit of radius r_2 . The gravitational force-distance graph is shown for the satellite.

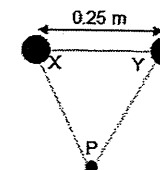


What does the shaded area on the graph represent?

- A The change in gravitational potential energy of the satellite
- B The change in kinetic energy of the satellite
- C The final gravitational potential energy of the satellite
- D The final kinetic energy of the satellite

11. [JJC/Prelims 2014/P1/20]

Two objects, each of mass 8000 kg and 0.25 m apart are placed at points X and Y as shown below. XYP forms an equilateral triangle.

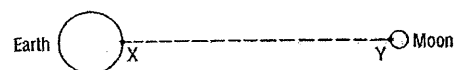


What is the gravitational potential energy of a mass of 5.0 kg placed at point P?

- A $-1.1 \times 10^{-5} \text{ J}$
- B $-2.1 \times 10^{-5} \text{ J}$
- C $-4.3 \times 10^{-5} \text{ J}$
- D $-8.5 \times 10^{-5} \text{ J}$

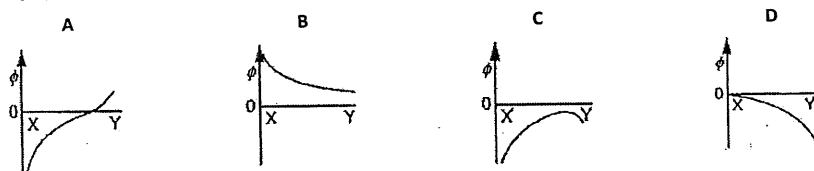
12. [JJC/Prelims 2014/P1/21]

The diagram below (not to scale) represents the relative positions of the Earth and the Moon.



The line XY joins the surface of the Earth to the surface of the Moon.

Which graph represents the variation of gravitational potential ϕ along the line XY?



13. [MJC/Prelims 014/P1/13]

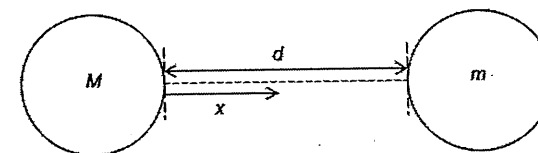
A satellite is moved from a low orbit to a higher orbit.

Which of the following correctly describes the energy of the satellite?

	total energy	gravitational potential energy	kinetic energy
A	stays the same	decreases	increases
B	stays the same	increases	decreases
C	increases	decreases	increases
D	increases	increases	decreases

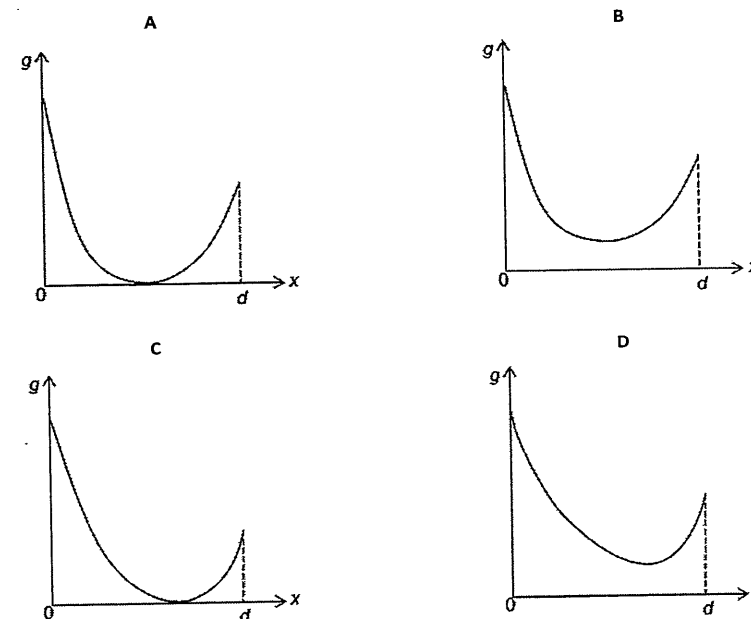
14. [MJC/Prelims 2014/P1/14]

Two isolated masses M and m of the same radius, are held a distance d apart, as shown below. Mass M is greater than mass m .



The gravitational field strength g is measured on a line between the two masses.

Which graph best shows the variation with distance x from the larger sphere of the magnitude of the field strength g ?



15. [NJC/Prelims 2014/P1/14]

A satellite skims the Moon's surface in a circular orbit. The acceleration at the Moon's surface due to gravity is $\frac{g}{6}$ and the Moon's radius is $\frac{R}{4}$, where g is the acceleration due to gravity on the surface of the Earth and R is the radius of the Earth.

If a satellite, skimming the Earth's surface has a period T , the period of revolution of the Moon's satellite will be

- A $\frac{2T}{3}$ B $\sqrt{\frac{2}{3}}T$ C $\left(\frac{2}{3}\right)^2 T$ D $\sqrt{\frac{3}{2}}T$

16. [NJC/Prelims 2014/P1/15]

A mass m is at fixed point Q. It produces a gravitational potential at point P, distant r from Q. This gravitational potential is equal to the external work done on unit mass in moving it

- A from P to Q B from Q to P C from P to infinity D from infinity to P

17. [NYJC/Prelims 2014/P1/13]

The average density of Planet P is twice that of Planet Q, and the radius of Planet P is half that of Planet Q. The gravitational field strength at the surface of P is 13.4 N kg^{-1} .

What is the gravitational field strength at the surface of Q?

- A 3.4 N kg^{-1} B 13.4 N kg^{-1} C 53.6 N kg^{-1} D 80.4 N kg^{-1}

18. [NYJC/Prelims 2014/P1/14]

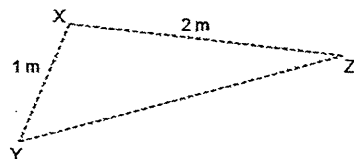
Which equation correctly shows the relationships between potential energy U , kinetic energy K and the total energy, E , of a satellite?

- A $U = -2K = 2E$ B $U = -K = E$ C $U = -\frac{1}{2}K = \frac{1}{2}E$ D $U = K = \frac{1}{2}E$

19. [PJC/Prelims 2014/P1/13]

Points X, Y and Z form a triangle as shown below. Line XY is 1 m long and line XZ is 2 m long, but line YZ is of unknown length.

A mass of $5.0 \times 10^3 \text{ kg}$ is placed at Y and two masses, each of $3.0 \times 10^3 \text{ kg}$, are placed at X and Z.



Which of the following are possible values of the gravitational potential energy and gravitational force acting on the mass at X?

	gravitational potential energy / 10^{-4} J	gravitational force / 10^{-4} N
A	-7.0	8.5
B	-13	12
C	-7.0	9.5
D	-13	9.2

20. [PJC/Prelims 2014/P1/14]

An external agent does 50 J of work in moving a mass of 3.0 kg from point A to point B, and -60 J of work in moving the mass from point B to point C. Finally, the external agent does work of 1000 J in moving the mass from point C to infinity.

What is the gravitational potential at point A?

- A -370 J kg^{-1} B -330 J kg^{-1} C 330 J kg^{-1} D 370 J kg^{-1}

21. [RVHS/Prelims 2014/P1/11]

Planet X and planet Y have the same mean density. The radius of planet X is half that of planet Y.

What is the ratio $\frac{\text{gravitational field at the surface of planet X}}{\text{gravitational field at the surface of planet Y}}$?

- A $2\sqrt{2}$ B 2 C $\sqrt{2}$ D $\frac{1}{2}$

22. [RVHS/Prelims 2014/P1/12]

Two point masses P and Q of masses M_P and M_Q are at a distance d apart.

What is the magnitude of the gravitational potential and the magnitude of the gravitational field strength caused by M_P at Q?

	gravitational potential	gravitational field strength
A	$-\frac{GM_P}{d}$	$\frac{GM_P M_Q}{d^2}$
B	$-\frac{GM_P}{d}$	$\frac{GM_P}{d^2}$
C	$-\frac{GM_Q}{d}$	$\frac{GM_Q}{d^2}$
D	$-\frac{GM_Q}{d}$	$\frac{GM_P M_Q}{d}$

23. [SAJC/Prelims 2014/P1/13]

Which quantity is *not necessarily* the same for satellites that are in geostationary orbits around the Earth?

- A kinetic energy
B angular velocity
C centripetal acceleration
D orbital period

24. [SRJC/Prelims 2014/P1/13]

Which of the following is/are factor/s that affects the energy input required for a rocket on surface of Mars to escape from the gravitational pull of Mars?

- I Mass of rocket
II Direction of launch with respect to direction of spin of Mars
III Geographic position of launch of rocket
IV Density of atmosphere of Mars

- A All of the above.
B II and IV
C I, III and IV
D None of the above.

25. [SRJC/Prelims 2014/P1/14]

Which of the following is the most accurate description of the changes in gravitational potential energy of a rocket as it escapes from the surface of Mars?

- A Gravitational potential energy decreases to zero.
- B The loss in kinetic energy of rocket is all converted into gravitational potential energy as it moves through the atmosphere.
- C Gravitational potential energy increases at a decreasing rate per unit distance.
- D Gravitational potential energy changes are always negative as gravitational forces are always attractive.

26. [TJC/Prelims 2014/P1/14]

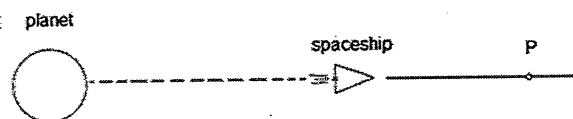
A spacecraft is in orbit at a distance r from the centre of the Earth. The engine of the spacecraft is fired and it moves to a new orbit of radius $2r$.

Which of the following describes the variations in angular speed and total energy of the spacecraft?

	angular speed	total energy
A	decreases	increases
B	decreases	decreases
C	increases	increases
D	increases	decreases

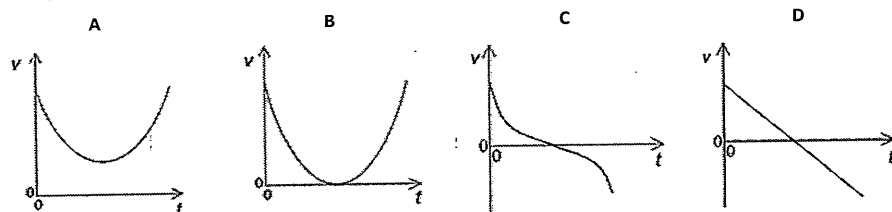
27. [TJC/Prelims 2014/P1/15]

A powered spaceship is moving directly away from a planet as shown below.



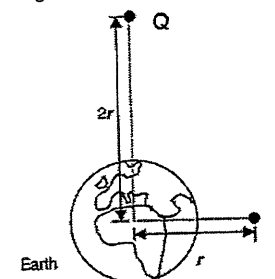
At point P, the motors of the spaceship are switched off but the spaceship remains under the influence of the planet.

Which graph best represents variation with time t of the velocity v of the spaceship after it passes point P?



28. [TPJC/Prelims 2014/P1/12]

P and Q are two points above the Earth's surface and at respective distances r and $2r$ from the centre of the Earth. The gravitational potential at P is -800 kJ kg^{-1} .

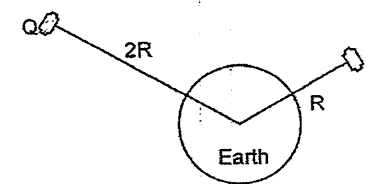


When a 1.00 kg mass is taken from P to Q, what is the work done on the mass by the external agent?

- A -400 kJ
- B -200 kJ
- C 400 kJ
- D 800 kJ

29. [TPJC/Prelims 2014/P1/13]

Two satellites, P and Q, of equal mass, orbit the Earth with radii as shown in the diagram.



Which statement is true?

- A P has lower speed and more potential energy than Q.
- B P has lower speed and less potential energy than Q.
- C P has higher speed and more potential energy than Q.
- D P has higher speed and less potential energy than Q.

30. [VJC/Prelims 2014/P1/13]

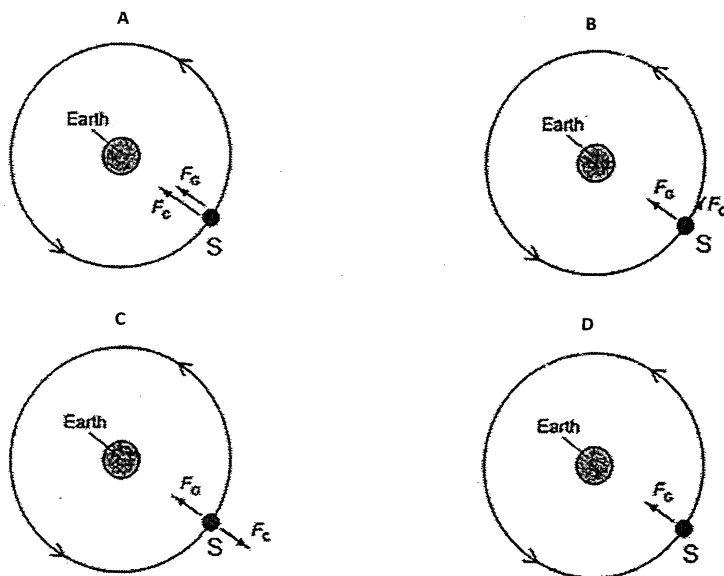
If a body of mass m were released in a vacuum just above the surface of a planet of mass M and radius R , what would be its gravitational acceleration?

- A $\frac{Gm}{R}$
- B $\frac{Gm}{R^2}$
- C $\frac{GM}{R}$
- D $\frac{GM}{R^2}$

31. [VJC/Prelims 2014/P1/14]

A satellite S orbits the Earth. The gravitational force on the satellite is F_g . The centripetal force required to maintain the satellite in orbit is F_c .

Which diagram shows the force, or forces, acting on the orbiting satellite?



32. [YJC/Prelims 2014/P1/13]

The difference in gravitational potential between a point on the surface of a planet X and a point 20 m above the surface is 6.0 J kg^{-1} .

Assuming that the gravitational field is uniform, how much work is needed to move a mass of 2.0 kg from the surface to a point 5.0 m above the surface?

- A 1.5 J B 3.0 J C 6.0 J D 12 J

33. [YJC/Prelims 2014/P1/14]

The Singapore Earth station is launching a new satellite of mass 320 kg into its orbit of radius $7.55 \times 10^6 \text{ m}$. The mass of Earth is $6.0 \times 10^{24} \text{ kg}$ and radius of Earth is 6370 km .

What is the speed of the satellite when it is moving in its orbit?

- A $3.46 \times 10^3 \text{ m s}^{-1}$ B $7.28 \times 10^3 \text{ m s}^{-1}$ C $1.03 \times 10^4 \text{ m s}^{-1}$ D $2.60 \times 10^4 \text{ m s}^{-1}$

II. Structured Questions

1. [AJC/Prelims 2014/P2/3]

The curve in Fig. 1.1 shows the way in which the gravitational potential energy of a body of mass m in the field of the Earth depends on r , the distance from the centre of the Earth, for values of r greater than the Earth's radius R_E .

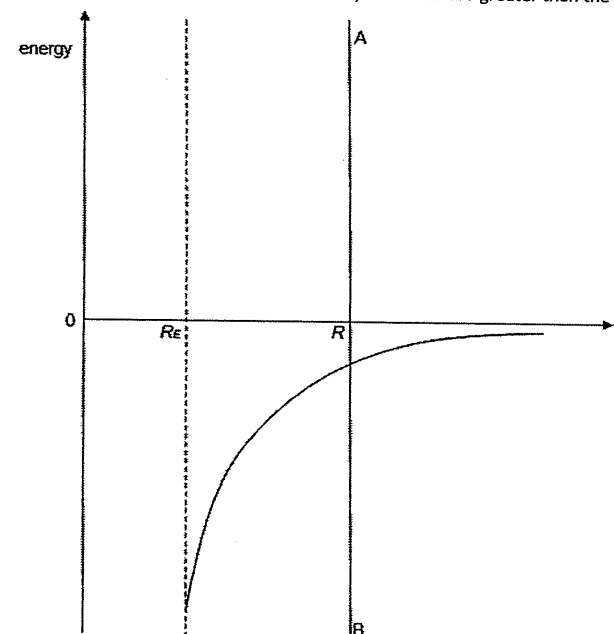


Fig. 1.1

The body referred in Fig. 1.1 is a rock which is projected vertically upwards from the Earth. Assume air resistance can be neglected.

At a certain distance R from the centre of the Earth, the total energy of the rock (i.e. its gravitational potential energy plus its kinetic energy) may be represented by a point on the line AB.

- For a rock that is momentarily at rest at the top of its trajectory, which is at the distance R from the centre of the Earth,
 - mark its total energy on the line AB in Fig. 1.1 with a cross and label it P. [1]
 - sketch the variation of its kinetic energy with distance from the centre of the Earth on Fig. 1.1. [2]
- The escape speed of the rock is the minimum speed that the rock must possess when it is on the Earth's surface so that it can escape to infinity.
 - On Fig. 1.1, mark on the line AB with a cross and label it Q to represent the total energy of the rock if it were projected from the Earth with the escape speed. [1]
 - Explain why the minimum speed for the rock to reach the moon from the surface of the Earth is less than the escape speed. [1]

[Total: 5]

2. [CJC/Prelims 2014/P3/1]

- a. State what is meant by gravitational potential at a point. [1]
- b. The variation of the gravitational potential between the Earth and the Moon with displacement from the centre of the Earth along the line joining their two centres is shown in Fig. 2.1 (not to scale) below.

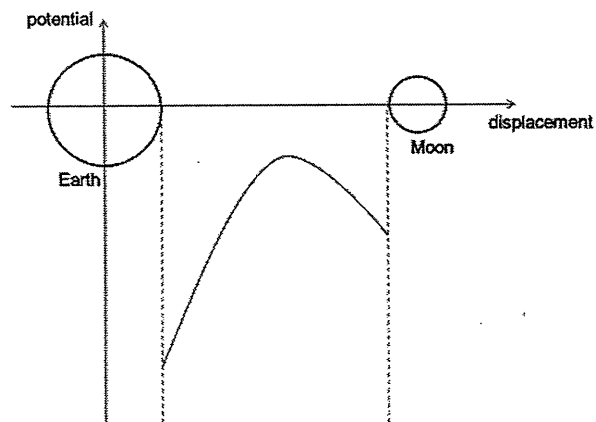


Fig. 2.1

On the displacement axis of Fig 2.1, label a point P at which an object will experience no resultant gravitational force. Explain your reasoning. [3]

- c. The point P in part (b), where an object experiences zero resultant force, is known as the neutral point.
- i. Let the distance from the centre of the Earth to point P be x , and the distance from the centre of the Moon to point P be y .

Show that the expression that relates x and y , to the masses of the Earth, M_E and the Moon, M_M is

$$\frac{x}{y} = \sqrt{\frac{M_E}{M_M}} \quad [2]$$

- ii. The mass of the Moon is 7.40×10^{22} kg, the mass of the Earth is 5.97×10^{24} kg, and the distance between the centres of the Earth and the Moon is 3.84×10^5 km. Calculate the distance from the centre of the Earth to point P. [2]
- d. Explain why it is not possible to place an object in a stable orbit around the Earth at the distance calculated in (c)(ii) from the centre of the Earth. [2]

[Total: 10]

3. [DHS/Prelims 2014/P3/6]

- a. Define gravitational field strength. [1]
- b. The mass M of a spherical planet may be assumed to be a point mass at the centre of the planet. A stone, travelling at speed v , is in a circular orbit of radius r about the planet, as illustrated in Fig. 3.1.

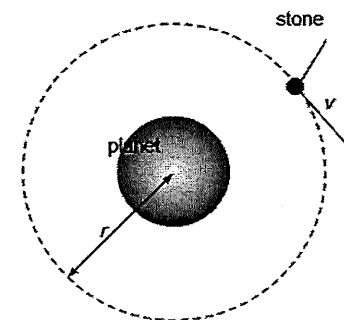


Fig. 3.1

Show that the speed v is given by the expression

$$v = \sqrt{\frac{GM}{r}} \quad [2]$$

- c. Define gravitational potential at a point. [1]
- d. Fig. 3.2 shows a second stone, initially at rest at infinity, traveling towards the planet.

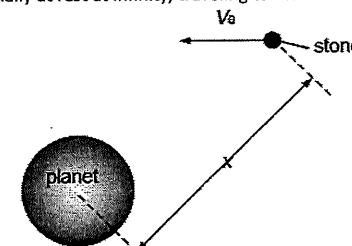


Fig. 3.2

The stone does not hit the surface of the planet.

- i. Determine, in terms of the gravitational constant G and the mass M of the planet, the speed V_0 of the stone at a distance x from the centre of the planet. Explain your working. You may assume that the gravitational attraction on the stone is due only to the planet. [3]
- ii. Use your answer in d. i. and the expression in b. to explain whether this stone could enter a circular orbit about the planet. [2]

- e. Fig. 3.3 shows a typical weather balloon consisting of a helium-filled lifting bag connected to a sensor package for measuring atmospheric parameters.

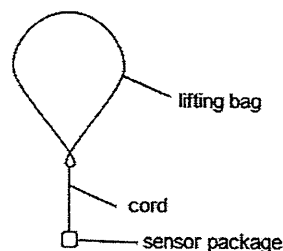


Fig. 3.3

It can reach a maximum height of 40 km above the ground.

- i. Explain why the lifting bag experiences a net upward force at the ground. [2]
- ii. Identify the force(s) acting on the weather balloon and state the relationship between the magnitudes of the forces when the weather balloon reaches maximum height. [1]
- iii. The radius of Earth, R_E , is 6400 km.
Show that the gravitational field strength, g , experienced by a weather balloon at maximum height deviates from that at the ground by 1.2%. Take g at the ground to be 9.81 m s^{-2} . [2]
- iv. The tension in the cord connecting a sensor package to the lifting bag hovering at maximum height was found to be 2.42 N.
Determine the mass of the sensor package. [1]
- v. The lifting bag bursts at the maximum height. An estimate of the time taken for the free fall of the sensor package to reach the ground, t , can be made by assuming that the gravitational field strength remains constant.
 1. Determine t using the value of gravitational field strength at the maximum height. [2]
 2. Explain whether your answer for t is an over-estimate or an under-estimate. [1]
- f. The lifting bag will burst at a fixed maximum volume.
Explain why increasing the initial volume of the lifting bag at ground level will cause the bag to burst at a lower maximum height. [2]

[Total: 20]

4. [HCI/Prelims 2014/P3/5]

- a. i. Define angular velocity for an object travelling in a circle. [1]
- ii. Calculate the angular velocity of the Earth in its orbit around the Sun. Assume that the orbit is circular. [1]
- b. A satellite with a mass 6200 kg is at point X, a distance of $1.50 \times 10^9 \text{ m}$ from the centre of the Earth, as shown in Fig. 4.1.

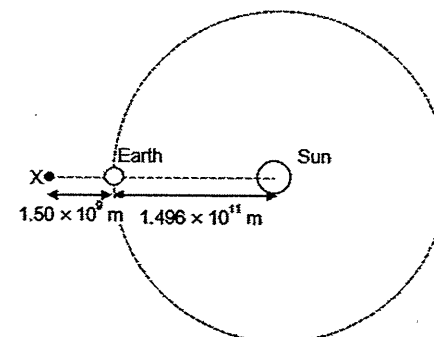


Fig. 4.1 (not to scale)

Mass of the Sun = $1.99 \times 10^{30} \text{ kg}$
 Mass of the Earth = $5.98 \times 10^{24} \text{ kg}$
 Earth-Sun distance = $1.496 \times 10^{11} \text{ m}$

- i. Show that the resultant force acting on the satellite due to the Sun and the Earth is 37.1 N. State the direction of this force. [4]
- ii. The satellite is in a circular orbit around the Sun.
Calculate the angular velocity of the satellite. [2]
- iii. Using your answer to a. ii. and b. ii., describe the motion of the satellite relative to the Earth. [1]
- c. A satellite P of mass m orbits the Earth in circular path as shown in Fig. 4.2. The satellite P has a speed v and the radius of its orbit is r .

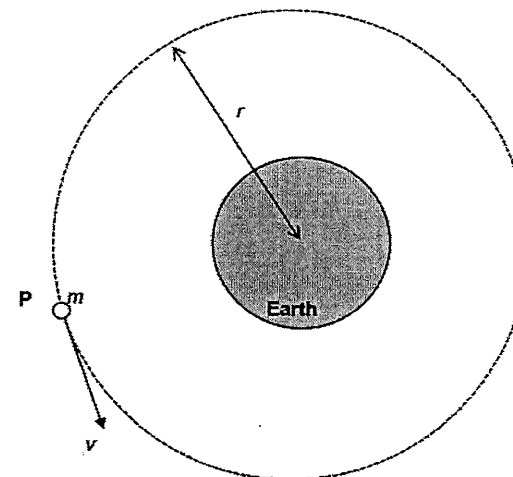


Fig. 4.2

- i. The speed v is related to the radius r by the relation $v = kr^n$.
Derive the expression for k and n in terms of G , M and m , where M is the mass of the Earth and G is the gravitational constant. [3]
- ii. Show that the total energy E_T of the satellite P is given by the expression

$$E_T = -\frac{GMm}{2r}$$
 [2]
- iii. The satellite P has a mass $m = 2200$ kg, and the orbital radius $r = 6.70 \times 10^6$ m. The radius of the Earth is 6.40×10^6 m. Calculate
- the minimum energy that must be supplied to satellite P to escape the Earth's gravitational field. [2]
 - minimum energy that must be supplied to satellite P to launch it from rest at the surface of the Earth into its circular orbit. [2]
- iv. If the satellite had a mass of 2.99×10^{24} kg, with the aid of a diagram, describe how the motion of the satellite differs from the one shown in Fig. 4.2 [2]

[Total: 20]

5. [JJC/Prelims 2014/P2/2]

- a. i. Define the terms gravitational field strength g and gravitational potential ϕ at a point in a gravitational field. [2]
- ii. Write down an expression that shows the relationship between the above two quantities with distance x from a point mass. [1]
- b. The values for the gravitational potential due to the Earth are given in the table below.

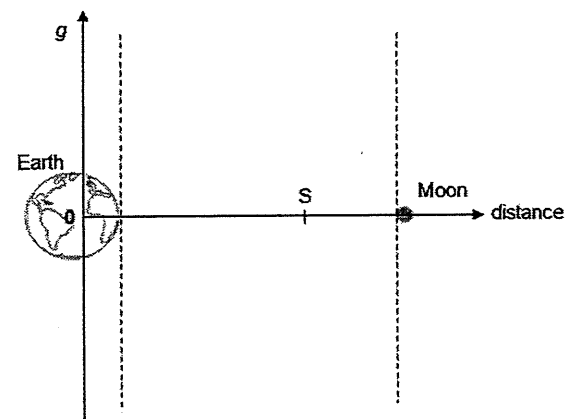
distance from Earth's surface / m	gravitational potential / MJ kg ⁻¹
0	-62.53
290 000	-59.82
300 000	-59.73
310 000	-59.64
∞	0

- i. A satellite of mass 900 kg falls from a height of 300 000 m to the Earth's surface.
Determine the potential energy it lose. [2]
- ii. Deduce a value for the Earth's gravitational field strength at a height of 300 000 m. [2]

[Total: 7]

6. [JJC/Prelims 2014/P3/7]

- a. State Newton's law of gravitation. [1]
- b. The mass M of the Earth may be considered to be concentrated at its centre. The radius of the Earth is R .
Show that the Earth's gravitational field strength on its surface is $g = \frac{GM}{R^2}$. [2]
- c. The distance from the centre of the Earth to the centre of the Moon is 3.8×10^8 m, the mass of the Earth is 6.0×10^{24} kg and the mass of the Moon is 7.0×10^{22} kg. A point S is located between the Earth and the Moon at a distance r from the centre of the Earth along the line joining the centres of the Earth and the Moon. Determine the value of r , if S is the point where the resultant gravitational field strength is zero. [3]
- d. On the axes below, sketch a graph (without any values) showing the variation of gravitational field strength g with distance from the Earth, along the line joining the surfaces of the Earth and the Moon. [2]



- e. Geostationary satellites are used in various applications such as weather forecasting and television broadcasting.
- State what is meant by geostationary. [1]
 - State two conditions for the satellite to be in geostationary orbit. [2]
- f. Singapore's first commercial remote sensing satellite, TeLEOS-1 is planned to be launched in 2015. It will have a mass of 400 kg and be designed to orbit above the equator at a height of 550 km. Take the radius of Earth to be 6.38×10^6 m and mass of Earth to be 5.97×10^{24} kg.
- Calculate the speed of TeLEOS-1 when in orbit. [2]
 - Calculate the total energy of TeLEOS-1 when in orbit. [3]
 - Determine quantitatively whether TeLEOS-1 could be in geostationary orbit. [3]
 - State one advantage of a satellite in geostationary orbit. [1]

[Total: 20]

Raffles Institution

Revision Package (H2 Physics)

8 – Gravitational Fields

7. [MJC/Prelims 2014/P2/2]

a.

An astronomer is investigating the relationship between the radius R and mass M of eight different stars of the same species. The graph of the ratios R/R_s against M/M_s of these eight different stars is plotted in Fig. 7.1, where R_s and M_s are the Sun's radius and mass respectively.

Fig. 7.1

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Revision Package (H2 Physics)

8 – Gravitational Fields

i.

On Fig. 7.1, draw the best fit line for the relationship between R/R_s and M/M_s for these stars.

ii.

Determine the equation for this linear relationship.

iii.

Estimate the maximum ratio of M/M_s for the mass of this species of star.

iv.

Suggest why your answer to a. iii. is only an estimate.

v.

Suggest and explain whether the Sun belongs to this species of stars.

b.

Fig. 7.2 shows the variation of the total gravitational potential with distance from the centre of one of the eight stars. R is the radius of the star which is 3300 km.

Distance from centre of the star	Gravitational potential / MJ kg ⁻¹
1.0 R	-13.5
1.5 R	-9.4
2.0 R	-7.2
2.5 R	-6.0
3.0 R	-5.2
3.5 R	-4.6
4.0 R	-4.2
4.5 R	-3.8
5.0 R	-3.5
5.5 R	-3.3
6.0 R	-3.0

Fig. 7.2

A rocket of mass 1.2×10^4 kg lifts off from the surface of this star.

i.

Use Fig. 7.2 to estimate the change in gravitational potential of the rocket at a distance $3R$ above the surface of the star

ii.

Use Fig. 7.2 to estimate the minimum launch speed that the rocket must have in order to reach a height of $3R$ above the surface of the star

iii.

Use Fig. 7.2 to estimate the magnitude of the gravitational field strength at a distance $3R$ above the surface of the star

iv.

Student A claims that the gravitational field strength at any point above the surface of this star is not inversely proportional to the square of the distance from the centre of this star.

Discuss the validity of this claim.

Total: 14

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8. [NYJC/Prelims 2014/P3/6]*

The Fig. 8.1 below represents three equipotential surfaces associated with the gravitational field of the planet Mars.

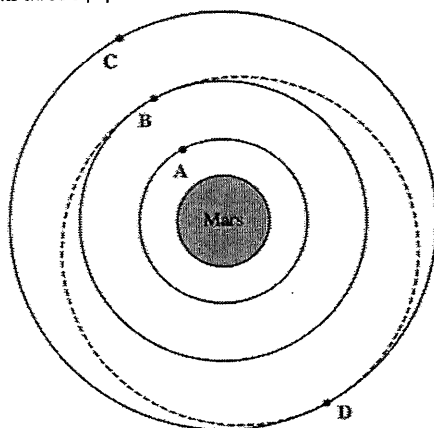


Fig. 8.1 (not to scale)

The points labelled A, B, C and D lie on the equipotential surfaces. The table in a. ii. shows the distances r of these equipotential surfaces from the centre of Mars and the gravitational potential ϕ for each surface. The gravitational potential at the surface of Mars is -12.6 MJ kg^{-1} . The dotted line shows the elliptical orbit that is referred to in part b. ii.

- a. i. State what is meant by the statement that the "gravitational potential at the surface of Mars is -12.6 MJ kg^{-1} ". [2]
- ii. Using the table below, determine the gravitational potential at point A. [2]

	r / km	$\phi / \text{MJ kg}^{-1}$
A	6800	
B	10200	-4.20
C	13600	-3.16

- b. i. An un-powered satellite in a stable orbit about its centre of Mars passes through C and D. Explain why the speed of the satellite is the same as D as it is at C. [1]
- ii. Another un-powered satellite of mass 850 kg passes point D at a speed of $1.64 \times 10^3 \text{ m s}^{-1}$. This satellite follows the elliptical orbit shown by the dotted line in Fig. 8.1. In this orbit the radius about the centre of Mars is continually changing. By considering the energy changes that occur, show the speed of this satellite when it passes point B is about $2.2 \times 10^3 \text{ m s}^{-1}$. [2]

- c. Mars has a mass of $6.4 \times 10^{23} \text{ kg}$ and a radius of $3.4 \times 10^6 \text{ m}$. The period of rotation of Mars about its axis is 24.6 hours.

Suppose that a future expedition lands on Mars and wishes to set up a communications satellite of mass 850 kg , in an orbit that has the same period as Mars.

- i. Calculate the radius of the orbit for this communication satellite. [2]
- ii. As the satellite orbits the Mars, it gradually loses energy due to small resistive forces. State and explain the effect of this change on the radius of the orbit and the speed of satellite. [3]
- d. A Simplified Aid for Extravehicular activity Rescue (a SAFER) is a device that astronauts have available in case they are accidentally detached from their safety line when performing tasks outside the space station. A SAFER is strapped to the astronaut's back and enables the astronaut to change direction and be propelled back to the space station. A SAFER uses a tank that initially contains 1.40 kg of liquid nitrogen which the SAFER converts to nitrogen gas under pressure. A thrust is produced as the gas expands and is released through one of the nozzles. Density of liquid nitrogen = 810 kg m^{-3}
Molar mass of nitrogen = 0.0280 kg
- i. Calculate the volume of liquid nitrogen that is carried by the SAFER. [1]
- ii. Calculate the volume of a container that 1.40 kg of nitrogen would occupy at a pressure of $1.0 \times 10^5 \text{ Pa}$ and a temperature of 25°C . Assume that the nitrogen behaves as an ideal gas. [2]
- iii. Use momentum considerations to explain how the release of gases from a nozzle propels the astronaut. [2]
- iv. Explain how the first law of thermodynamics applies when the gas is under pressure in the SAFER expands through a nozzle. [3]

[Total: 20]

9. [NYJC/Prelims 2014/P3/1]

- a. i. Define gravitational potential. [2]
- ii. Explain why gravitational potential has a negative value. [2]
- b. Values for the gravitational potential due to the Earth are given in the table below.

distance from Earth's surface / m	gravitational potential / MJ kg^{-1}
0	-62.72
390 000	-59.12
400 000	-59.03
410 000	-58.94
∞	0

- i. Calculate the loss in the potential energy of a satellite of mass 700 kg if it falls from a height of $400\,000 \text{ m}$ to the surface of the Earth. [2]
- ii. Deduce a value for the gravitational field strength of the Earth at a height of $400\,000 \text{ m}$. [2]

[Total: 8]

10. [PJC/Prelims 2014/P3/6]

- a. Write down an equation expressing Newton's law of gravitation. Define all symbols. [3]
- b. A geostationary satellite has to be placed above the equator.
- State what is meant by geostationary satellite. [1]
 - State the orbital period and direction of rotation of such a satellite. [1]
 - Explain why the satellite must be above the equator. [2]
- c. Fig. 10.1 shows a geostationary satellite in orbit above the Earth. The satellite sends out a single microwave pulse to a reflector placed vertically below it and receives the reflected signal.

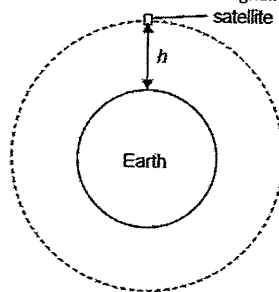


Fig. 10.1

The Earth has an approximate mass of 5.97×10^{24} kg and an average radius of 6371 km while the geostationary satellite has a mass of 100 kg.

- Calculate the height, h of the satellite above the Earth. [4]
 - Hence, calculate the time it takes for the pulse to travel from the satellite to the Earth and back to the satellite. [2]
 - Explain if your answer to c. i. will increase, decrease or remain the same if the mass of the satellite is doubled. [2]
 - Calculate the gravitational force acting on the satellite. [2]
- d. Fig. 10.2. shows a moment in time when the Moon is at an angle θ to the line joining the satellite and the Earth.

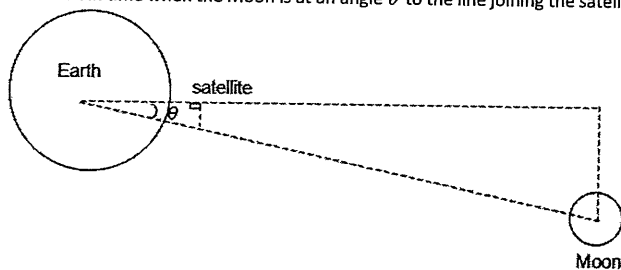


Fig. 10.2

Explain qualitatively why geostationary satellites need to be equipped with thrusters which fire periodically to maintain their orbits. [3]

[Total: 20]

11. [RVHS/Prelims 2014/P3/7]

- a. A space shuttle is currently between Saturn and Jupiter as shown in Fig. 11.1.

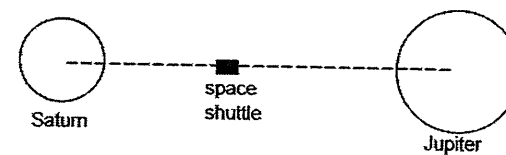


Fig. 11.1

Its mission is to obtain information regarding the surface of Saturn.

Mass of space shuttle = 2.2×10^3 kg

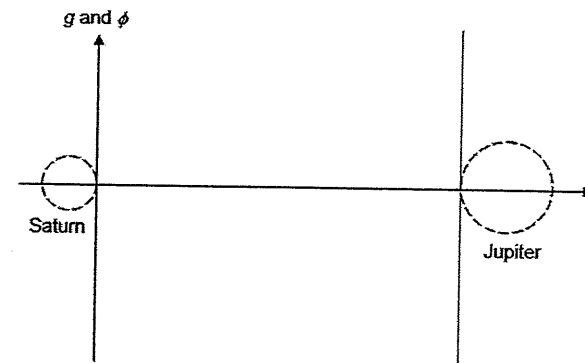
Mass of Saturn = 5.7×10^{26} kg

Mass of Jupiter = 1.9×10^{27} kg

One Saturn day = 10.7 hours

One Saturn day refers to the time it takes for Saturn to complete one revolution about its own axis.

- Calculate the resultant gravitational field strength experienced by the space shuttle when it is at a distance 3.9×10^{11} m from Saturn and 4.2×10^{11} m from Jupiter. [2]
- Sketch a graph below to represent the variation of the resultant gravitational field strength g with the distance r from the surface of Saturn to the surface of Jupiter. Label your graph as ii. Take rightwards to be positive. (Numerical values for g are not expected). [2]



- Indicate with a cross on your graph the position of the space shuttle as stated in i. [1]
- On the same axes as ii., sketch a graph to represent the total gravitational potential between the surfaces of Saturn and Jupiter. Label your graph as iv. [2]
- When the space shuttle arrives in the vicinity of Saturn, it is set into a free orbit around Saturn, such that it remains above the same point on Saturn's equator throughout one complete Saturn day. This is called a "stationary" orbit.

For computation regarding this stationary orbit, you may ignore the effect of Jupiter's gravitational field.

Calculate

- the period of this orbit in seconds [1]
- the radius of this stationary orbit. [2]

- vi. Explain why the space shuttle must be placed above the Saturn's equator in order for it to be in a stationary orbit. [1]
- vii. When the space shuttle is in this orbit, an object of mass 2.5 kg is placed on a spring balance that is attached to the inside of the space shuttle. State and explain the reading shown on the spring balance. [2]
- b. The space shuttle is sent on another mission to investigate another planet that has several moons orbiting around it. Data for some of the moons of the planet are given in Fig. 11.2.

moon	radius of orbit / m	period / days
A	1.81×10^8	0.500
B	6.71×10^8	3.55
C	1.88×10^9	16.7
D	2.12×10^{10}	631

Fig. 11.2

- i. Use any two sets of readings from Fig 11.2 to verify that T^2 is directly proportional to R^3 , where T is the period of the moon around the planet and R is the radius of orbit of the moon around the planet. [2]
- ii. Hence state the constant of proportionality, with the correct units, for this relationship. [1]
- iii. It can be shown that the period of the moon around the planet and the radius of orbit is given by the equation $T^2 = \frac{4\pi^2 R^3}{g_p R_p^2}$, where g_p is the gravitational field strength on the planet's surface and R_p is the radius of the planet. Given that $R_p = 6.9 \times 10^7$ m, use b. ii. to estimate the value of g_p . [2]
- iv. By comparison, the Earth only has one moon orbiting around it. The distance between the Moon and the Earth is 3.84×10^8 m. Explain whether data from Fig. 11.2 can be used to calculate the period of our Moon around the Earth. [2]

[Total: 20]

12. [SAJC/Prelims 2014/P3/2]

- a. State what is meant by a geostationary satellite. [1]
- b. Explain why a satellite is normally launched near the equator. [1]
- c. A satellite of mass 120 kg is orbiting the Earth with an orbital radius, 4.23×10^7 m. It gradually moves to a lower orbit of radius, 3.00×10^7 m. The mass of the Earth is 5.98×10^{24} kg. Calculate for the satellite,
- the change in kinetic energy [2]
 - the change in gravitational potential energy. [2]
- d. Using energy considerations, explain why the satellite in c. increases in speed despite losing energy due to air resistance. [1]
- e. Explain why an astronaut in a satellite orbiting the Earth experiences "weightlessness" although the gravitational field strength at that height is not zero. [1]

[Total: 8]

13. [SRJC/Prelims 2014/P3/2]

- a. The relationship between a satellite's period of orbit T around a planet and the radius of the orbit r can be described by Kepler's 3rd Law as $T^2 = kr^3$, where k is a constant.
- State a physical quantity that affects the value of k . [1]
 - Based on Kepler's 3rd Law or otherwise, determine whether a satellite of mass m at an orbital path with a radius r has a higher or lower kinetic energy than a satellite of the same mass with a larger orbital radius of r' . [3]
- b. A satellite of Earth has a period that is the same as that of the moon of Earth. It lies along the straight line joining the centre of Earth and its moon. The period of the moon is 27.3 days. The graph of the resultant field strength of the Earth and its moon along the line joining the centres varies as shown in Fig. 13.1.

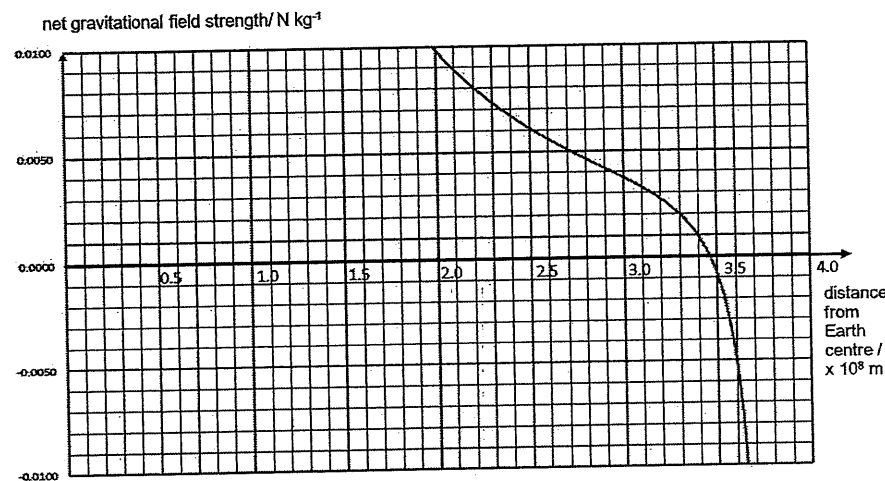


Fig. 13.1

- Derive the relationship between the gravitational field strength of the Earth g_e , gravitational field strength of the moon g_m and the period T of the satellite. [1]
- On Fig. 13.1, sketch a graph that will help determine the radius of the orbital path of the satellite [1]
- Describe and explain how the graph in b. ii. can be used to determine the radius of the orbital path of the satellite. [1]

[Total: 7]

14. [TJC/Prelims 2014/P3/5]

- a. State what is meant by a gravitational field. [1]
- b. The table shows how the gravitational potential varies at three points above the centre of an unknown planet.

distance from centre of planet / 10^5 m	gravitational potential / 10^3 J kg $^{-1}$
1.25 (surface of planet)	-15.0
3.00	-6.25
7.50	-2.50

- i. Show, using the data, that the potential is inversely proportional to the distance from the centre of the planet. [2]
- ii. Use the data to determine the gravitational field strength at the surface of the planet. [2]
- iii. It is suggested that a space vehicle could land on the planet to search for valuable minerals. The planet is spinning and the time for one rotation is 5.0 hours.
The space vehicle of mass 3.0×10^4 kg is located on the equator of the planet at a distance of 1.25×10^5 m from the centre, as shown in Fig. 14.1.

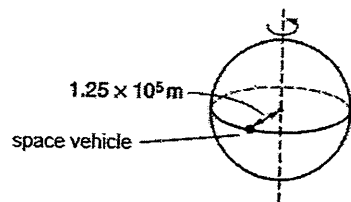


Fig. 14.1

1. Show that the centripetal force needed to keep the vehicle on the surface of the planet is 460 N. [2]
2. Hence deduce quantitatively whether the vehicle will stay on the surface of the rotating planet. [3]
- b. A space probe is launched from the north pole of the planet. During the launch, the energy E given to the space probe of mass m is

$$E = \frac{3GMm}{4R}$$

where G is the gravitational constant and M and R are the mass and radius of the planet respectively. Work done in overcoming frictional force can be neglected.

- i. Explain what is meant by escape speed. [1]
- ii. State and explain whether the space probe will be able to escape into deep space. [3]
- iii. The space probe is now launched into a circular polar orbit of radius r . Derive expressions, in terms of G , M , R , m and r for
- the change in gravitational potential energy of the space probe as it travels from the planet's surface to its orbit. [1]
 - the kinetic energy of the space probe when it is in its orbit. [2]
- iv. Use your answers in c. iii. and the total energy supplied to the space probe as given in c., to determine the height of the orbit above the planet's surface in terms of R . [3]

[Total: 20]

15. [TPJC/Prelims 2014/P3/6]

- a. i. Define angular velocity for an object travelling in a circle. [1]
- ii. Calculate the angular velocity of the Earth in its orbit around the Sun. Assume that the orbit is circular. [2]
- b. In order to observe the Sun continuously, a satellite of mass 425 kg is at point X, a distance of 1.60×10^9 m from the centre of the Earth, as shown in Fig. 15.1.

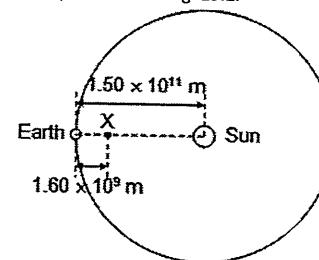


Fig. 15.1

Mass of the Sun = 1.99×10^{30} kg
 Mass of the Earth = 5.98×10^{24} kg
 Earth-Sun distance = 1.50×10^{11} m

- i. Calculate
- the magnitude and direction of the resultant force on the satellite [4]
 - the acceleration of the satellite. [2]
- ii. Assuming that the satellite is in a circular orbit around the Sun, calculate the angular velocity of the satellite. [2]
- iii. Using your answer to a. ii. describe the motion of the satellite relative to the Earth. [1]
- iv. Suggest why such an orbit around the Sun is preferred over a satellite orbit around the Earth. [1]
- v. Suggest a disadvantage of having a satellite in this orbit. [1]
- c. The orbital path around the Sun for the satellite in (b) is in fact not perfectly circular, but elliptical, as shown in Fig. 15.2.

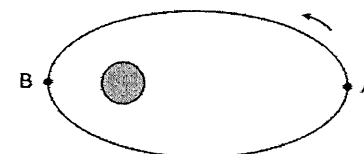


Fig. 15.2

- i. On Fig. 15.2, draw vectors representing
- the velocity of the spacecraft when it is at A and B (label them v_A and v_B) [1]
 - the acceleration of the spacecraft when it is at A and B (label them a_A and a_B) [1]
- ii. State and explain whether you have drawn
- the velocity vector at A longer, shorter or equal with respect to the velocity vector at B. [2]
 - the acceleration vector at A longer, shorter or equal with respect to the acceleration vector at B. [2]

[Total: 20]

16. [YJC/Prelims 2014/P3/71]*

- a. Define gravitational field strength. [1]
- b. Two small metal spheres, A and B of different masses, 100 kg and 400 kg with charges, +10 C and -5.0 C, respectively, are held in place so that the distance between the centres of spheres is kept at 60 m, as shown in Fig. 16.1.

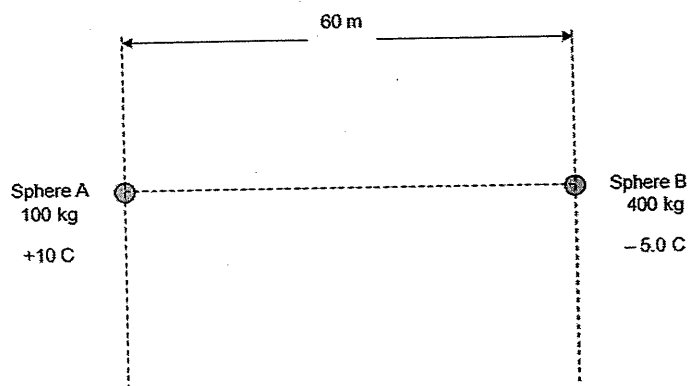


Fig. 16.1

- i. Determine the distance x from the centre of sphere A along the line joining A and B, where the net gravitational field strength, $Net\ g = 0$. [2]
- ii. On Fig. 16.2, show the variation with distance of the $Net\ g$ by sphere A and B. [4]

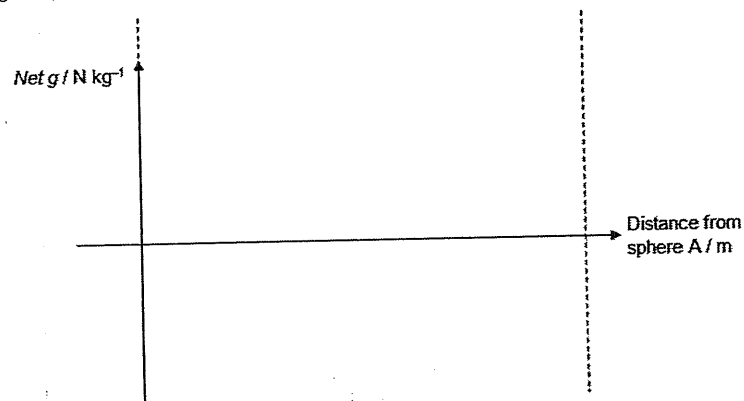


Fig. 16.2

- iii. State how gravitational field strength g at a point is related to gravitational potential ϕ at that point. [1]

- iv. Hence or otherwise, on Fig. 16.3, show the variation with distance of the $Net\ \phi$ of sphere A and B. [2]

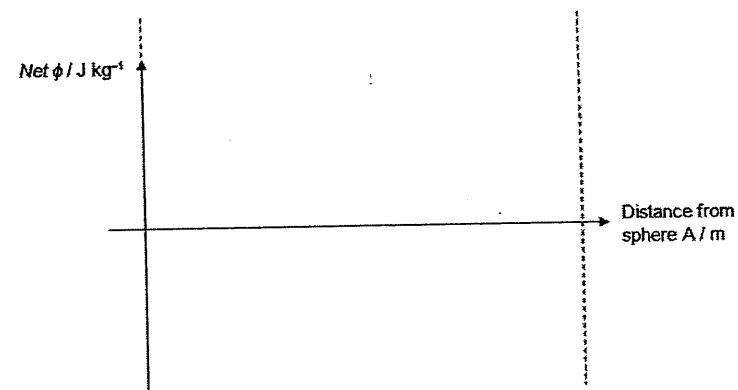


Fig. 16.3

- v. On Fig. 16.1, draw the resultant electric field lines around sphere A and B due to their charges. [2]
- vi. Calculate the magnitude of the net electric field strength at distance x , where $g = 0$. [2]
- vii. Briefly state and explain the subsequent motion of a small sphere C with mass 1 kg and charge +1 C, when it is placed at distance x . [2]
- viii. Define electric potential. [1]
- ix. Determine the distance y from the centre of sphere A along the line joining A and B, where the net electric potential of sphere A and B, $Net\ V = 0$. [2]
- x. Hence, on Fig. 16.4, show the variation with distance of the $Net\ V$ of sphere A and B. [2]

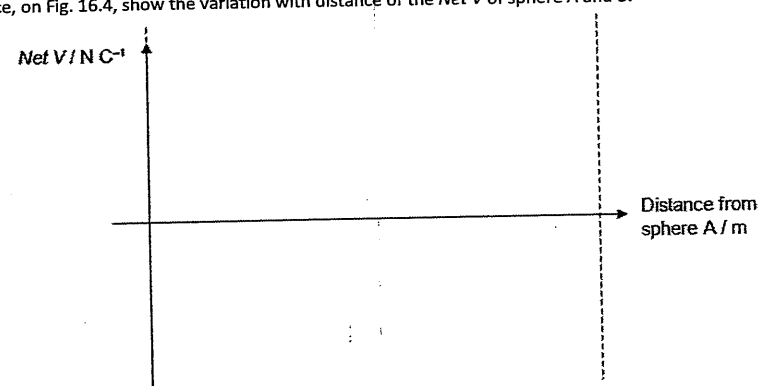


Fig. 16.4

- xi. Briefly state and explain whether the net electric field strength is zero at distance y , as calculated in part ix. [1]

[Total: 8]

III. Solutions and Answers

The suggested solutions and answers below are only for your reference and have not been verified to be error-free. If you have any doubt, please discuss with your tutor to clarify.

MCQs

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
B	A	A	D	D	A	B	D	B	A	B	C	D	C	D

16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
D	B	A	D	B	D	B	A	A	C	A	C	C	D	D

31	32	33
D	B	B

- 1 B At the point where gravitational force on mass m is zero the gravitational force on the mass m exerted by earth is equal and opposite to that exerted by moon on mass m . Applying Newton's law of gravitation to determine magnitude of these two forces

- 2 A Work done by external agent on mass = change in GPE of the 100g mass
 $\Delta U = m(\phi_f - \phi_i) = 0.100 \{ -2000 - (-2500) \} = 50 \text{ J}$

- 3 A $GPE_i + 0 = GPE_f + KE_f$
 $KE_f = GPE_i - GPE_f = - (GPE_f - GPE_i)$

$$\begin{aligned} \frac{1}{2}mv^2 &= - \left[\left(-\frac{GMm}{r_f} \right) - \left(-\frac{GMm}{r_i} \right) \right] = GMm \left(\frac{1}{r_f} - \frac{1}{r_i} \right) \\ v &= \sqrt{2GM \left(\frac{1}{r_f} - \frac{1}{r_i} \right)} \\ &= \sqrt{2(6.67 \times 10^{-11})(6.0 \times 10^{24}) \left(\frac{1}{6400 \times 10^3} - \frac{1}{(6400 + 620) \times 10^3} \right)} \\ &= 3.3 \times 10^3 \text{ m s}^{-1} \end{aligned}$$

- 4 D Resultant potential at the point:

$$\begin{aligned} \phi_{res} &= \phi_S + \phi_J \\ &= -\frac{GM_S}{r_S} + \left(-\frac{GM_J}{r_J} \right) \\ &= - \left(\frac{6.67 \times 10^{-11} \times 1.99 \times 10^{30}}{7.80 \times 10^{11}} \right) + \left(-\frac{6.67 \times 10^{-11} \times 1.90 \times 10^{27}}{3.49 \times 10^8} \right) \\ &= -5.33 \times 10^8 \text{ J kg}^{-1} \end{aligned}$$

Escape velocity: $\frac{1}{2}mv_E^2 = m|(\phi_S + \phi_J)|$

$$v_E = \sqrt{2|(\phi_S + \phi_J)|} = \sqrt{2(5.33 \times 10^8)} = 32.6 \text{ km s}^{-1}$$

- 5 D Gravitational force provides for centripetal force. $\frac{GMm}{r_g^2} = mr_g\omega_g^2 \Rightarrow \omega_g^2 = \frac{GM}{r_g^3}$

Let height of polar satellite be r_p , $r_p = R_E + \frac{r_g - R_E}{16}$

Similarly for the polar-orbiting satellite: $\omega_p^2 = \frac{GM}{r_p^3}$

$$\frac{\omega_p^2}{\omega_g^2} = \frac{GM}{r_p^3} \cdot \frac{r_g^3}{GM} = \frac{r_g^3}{r_p^3} = \frac{r_g^3}{(R_E + \frac{r_g - R_E}{16})^3} = \left(\frac{16r_g}{15R_E + r_g} \right)^3$$

$$\frac{\omega_p}{\omega_g} = \sqrt{\left(\frac{16r_g}{15R_E + r_g} \right)^3} = 64 \left(\frac{r_g}{15R_E + r_g} \right)^{3/2} = 64 \left(\frac{r_g}{1 + 15R_E/r_g} \right)^{3/2}$$

- 6 A Since ϕ becomes more negative $\Rightarrow$ satellite moves closer to Earth.

$$\begin{aligned} \Delta\phi &= \phi_f - \phi_i = -60 - (-20) \text{ MJ kg}^{-1} = -40 \text{ MJ kg}^{-1} \\ \Delta U &= m\Delta\phi = (50)(-40 \times 10^6) = -2000 \text{ MJ} \end{aligned}$$

- 7 B Projectile requires enough kinetic energy to project it till it reaches point P, after which it will be naturally attracted towards the Moon.

Considering from Earth till point P, by conservation of energy,
 Gain in GPE = Loss in KE

$$\begin{aligned} m(\phi_i - \phi_f) &= \frac{1}{2}mv^2 \\ [-1.3 - (-62.3)] \times 10^6 &= \frac{1}{2}v^2 \\ v &= 11.045 \times 10^3 \text{ m s}^{-1} \end{aligned}$$